OpenClaw Voice Dialogue Interaction

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1. Course Content

Course Overview

This lesson is a hands-on tutorial explaining how to use the voice dialogue interaction feature of the OpenClaw framework on the ROSMASTER-M1PRO robot.

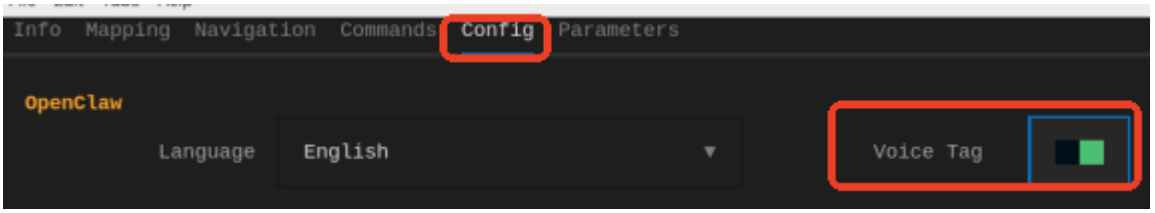
Learning Objectives

- Master how to enable and disable the OpenClaw voice reply tag
- Master the steps to configure the voice model parameter file and API-KEY
- Master the process of starting the MCP service and the multi_brains_pre voice node program
- Master how to conduct voice dialogue using the wake-up word

2. Enabling and Disabling the Voice Tag

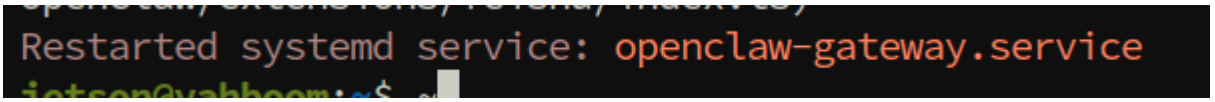
> **Note**
>
> - The OpenClaw voice reply tag is disabled by default at the factory. When you need to use the voice feature, enable and disable it through RobotManager

- Enable the voice tag (for the Raspberry Pi motherboard, also run it in the host computer terminal)

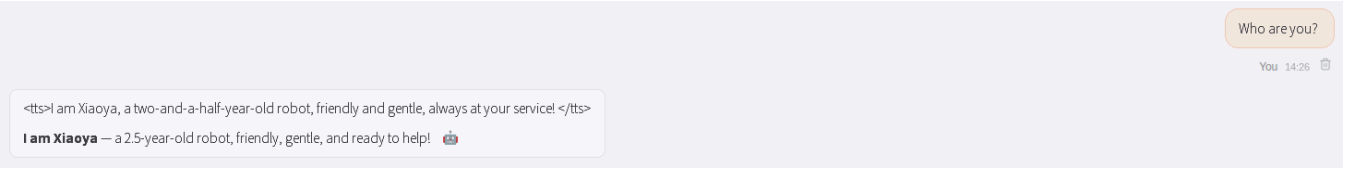


- Restart the OpenClaw gateway to apply the configuration

```
openclaw gateway restart
```



- Test the reply
  - Here we test on the web side. First use the `/clear` command to clear the dialogue history to avoid interference from the previously loaded old AGENTS.md, and enter a test sentence
  - You can see that the reply contains the `<tts>...</tts>` tag. OpenClaw's `reply-bridge` plugin automatically parses this tag and then calls the MCP interface for speech synthesis and broadcasting



3. Configure the Online Voice Model API-KEY (Optional)

```
robot_manage gui
```

- The factory default uses a local offline streaming speech recognition engine and local offline speech synthesis

> **Tip**
>
> - Configuration is only needed when using online speech recognition or online speech synthesis. For Raspberry Pi 5 (4GB) and RDK X5 motherboard users, it is recommended to **prioritize online speech recognition and speech synthesis** in the comprehensive feature cases involving SLAM navigation in the subsequent OpenClaw chapters, as this effectively reduces the CPU and memory pressure on the motherboard; in the single-instance OpenClaw chapters, you can still use the offline voice feature to save costs
> - Comparison of offline speech recognition and speech synthesis features

Feature	Solution	Effect Comparison
Offline Speech Recognition (ASR)	Local streaming speech recognition engine	No extra cost, real-time streaming recognition, but it uses some of the motherboard's CPU and memory resources
Online Speech Recognition (ASR)	Alibaba Cloud Bailian (Chinese) / iFlytek (English)	No streaming recognition, does not use the motherboard's computing resources
Offline Speech Synthesis (TTS)	Local speech synthesis engine	No extra cost; the synthesized voice is mechanical with a single timbre

Feature	Solution	Effect Comparison
Online Speech Synthesis (TTS)	Alibaba Cloud Bailian (Chinese) / iFlytek (English)	Better voice quality, does not use the motherboard's computing resources; the timbre is lifelike and close to a real person, with multiple timbres to choose from

Chinese Online Voice Settings

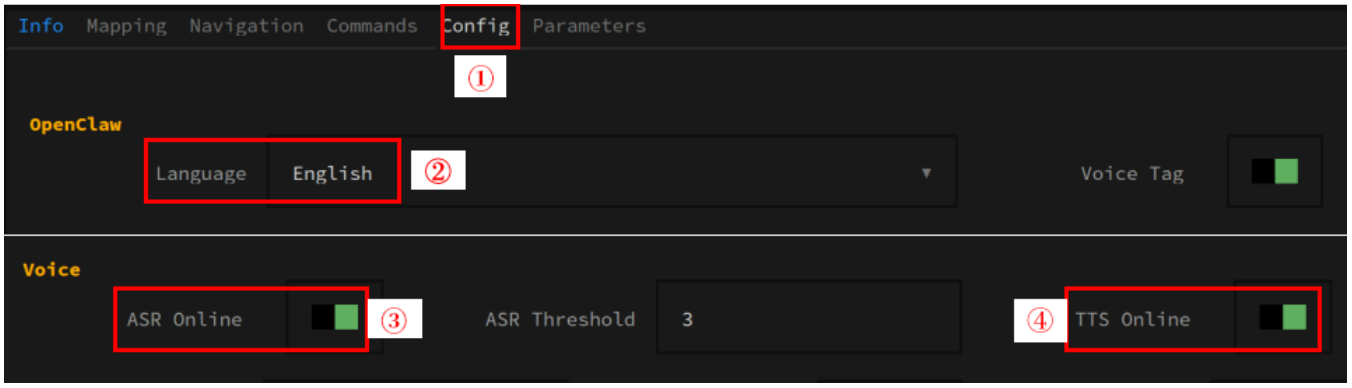
- Parameters for the China region use Alibaba Cloud Bailian speech synthesis by default; fill in the Alibaba Cloud Bailian API-KEY and baseURL





English Online Voice Settings

- For international regions, the English online speech recognition and speech synthesis are provided by iFlytek; no API-KEY configuration is needed, just switch the language to English



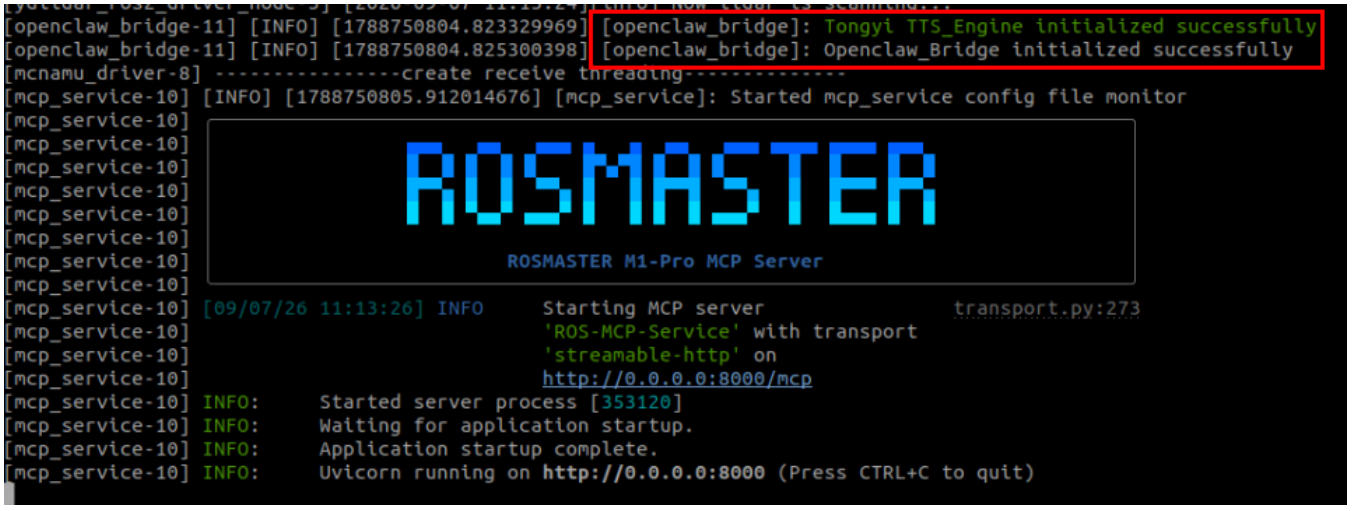
Offline Voice Settings

- Turn off the online ASR and online TTS switches to switch to offline speech recognition and speech synthesis



4. Preparation

```
ros2 launch yahboomcar_bringup mcp_service.launch.py use_tts:=true
```



- Start the speech recognition node

```
ros2 run multi_brains_pre asr_detect
```

- After startup, the log will indicate the loaded speech recognition model; the factory default uses the offline streaming speech recognition engine

```
jetson@yahboom:~$ ros2 run multi_brains_pre asr_detect
[kws_mic] Serial /dev/mic opened successfully
[voice_toolbox] v1.0.0 loaded, copyright Yahboom Technology
[INFO] [1777367265.037545660] [asr_detect]:  ASR Engine offline-stream initialized successfully
[INFO] [1777367265.077128639] [asr_detect]: VAD engine initialized successfully
[✓]  How is the weather today?
```

> **Important**
>
> - In the subsequent chapters, the voice playback feature is enabled through `use_tts:=true`, and the robot-side speech recognition requires starting the **asr_detect** node, started as needed. If you do not need the robot-side voice feature, you can omit the use_tts parameter and not start the voice node. This will not be repeated in the subsequent OpenClaw chapter lessons.

5. Voice Dialogue Case Demonstration

- Wake up the robot with the wake-up word "Hi Yahboom", and you can start the conversation
- If the local speech recognition engine is used, streaming speech recognition is performed

```
jetson@yahboom:~$ ros2 run multi_brains_pre asr_detect
[INFO] [1789197936.100555585] [asr_detect]: enable_wakeup_voice: True, enable_keybord_wakeup: False
[kws_mic] Serial /dev/myspeech opened successfully
[voice_toolbox] v1.0.0 loaded, copyright Yahboom Technology
[INFO] [1789197943.945509453] [asr_detect]:  ASR Engine offline-stream initialized successfully
[INFO] [1789197943.972691732] [asr_detect]: VAD engine initialized successfully
[✓]  Tell me a joke
```

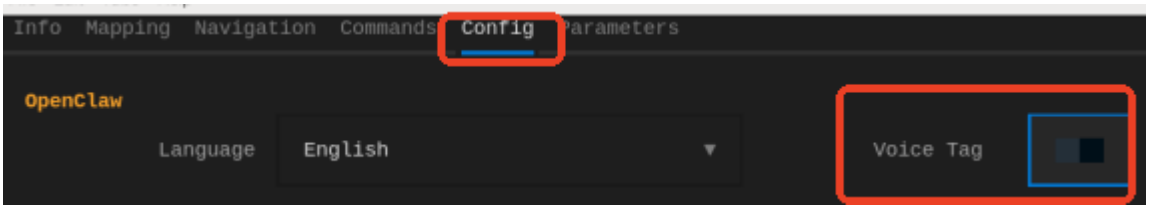
- The openclaw_bridge node displays OpenClaw's text reply and performs speech synthesis and broadcasting of the content within the voice tag

```
[mcp_service-10] INFO:     127.0.0.1:37836 - "POST /mcp HTTP/1.1" 200 OK
[openclaw_bridge-11] [INFO] [1789197760.904802495] [openclaw_bridge]: TTS request: Here's a programmer joke: Why can't
programmers tell Halloween from Christmas? Because Oct 31 equals Dec 25. Octal 31 equals decimal 25!
```

6. Function Q&A

1. If I do not want OpenClaw to output the voice reply tag, how do I disable it?

Answer: Disable the voice reply tag, restart the OpenClaw gateway, and ask OpenClaw in the conversation not to reply with voice:



2. Why does OpenClaw reply with a lot of text but only a very short voice reply?

Answer: The voice reply is a condensed version of the text, replying only with the key information. The robot only broadcasts the text within the `<tts>...</tts>` tag. The design reasons are as follows:

- Reduces the time needed for speech synthesis; overly long text greatly increases the time required for speech synthesis
- Overly long speech synthesis increases the token consumption cost of the voice model
- In most scenarios only a short voice reply is needed, and the text reply is already detailed enough

3. The OpenClaw reply has no voice tag and no voice reply was given. How can I have the full content replied by voice?

- You can require OpenClaw to always reply with voice during the conversation
- If you need the entire full content replied by voice, you can ask OpenClaw to do so during the conversation